IRT-FIT: Characterizing the Detection Frontier for Benchmark Contamination via Person-Fit Statistics

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Abstract

Benchmark contamination threatens the reliability of LLM evaluation, yet all existing detection methods require privileged access beyond binary correct/incorrect responses—token probabilities, model weights, or specially crafted re-tests. We establish the *minimal-information operating point* on the contamination detection frontier by repurposing IRT person-fit statistics (l_z^*), originally developed for detecting exam cheating. Through simulation, controlled contamination experiments across four models and three benchmarks (72 runs), and a large-scale audit of 6,200+ Open LLM Leaderboard models, we make three contributions: (1) We show that detection power is a predictable function of benchmark size, following a $\sqrt{n}$ relationship ($R^2 = 0.96$), achieving AUROC 0.78 on MMLU (14K items) at 15% contamination but degrading sharply for smaller benchmarks—precisely quantifying the cost of operating at the binary-only information level. (2) We derive a power function that predicts detection thresholds: $n \geq 8{,}000$ items are required for 80% power at 15% contamination, providing actionable guidance for benchmark design. (3) We conduct the first retrospective person-fit audit of 6,200+ models, revealing a significant temporal increase in anomaly rates (+3.1pp/quarter on MMLU, $p = 0.008$) and identifying 42 models with cross-benchmark anomalies. We openly characterize limitations: IRT-FIT trails richer-information methods by 4–18pp AUROC, and post-RL training eliminates the contamination signal (retention 19%). These findings position IRT-FIT as the only method capable of retrospective leaderboard-scale auditing, while precisely delineating when and where it succeeds.

1 Introduction

As LLM training corpora expand to encompass increasingly large fractions of the internet, benchmark contamination—where evaluation items appear in training data—has become a critical threat to evaluation integrity. Recent studies demonstrate that contamination inflates scores on standard benchmarks, undermining both model comparisons and scientific conclusions drawn from benchmark performance [Magar and Schwartz, 2022, Sainz et al., 2023, Jacovi et al., 2023].

Existing contamination detection methods span a spectrum of information requirements. N-gram overlap methods require access to training data [Brown et al., 2020]. Membership inference attacks (MIA) and Min-K%++ operate on token-level log-probabilities [Shi et al., 2024]. Time Travel probing requires specially designed prompts [Dong et al., 2024]. ConStat demands live model re-evaluation on rephrased items [Li et al., 2024]. Critically, *none* of these methods can operate on binary response patterns alone—yet binary correct/incorrect records are the most widely available evaluation data, preserved in leaderboards and evaluation logs for thousands of models.

We observe that a contaminated LLM answering benchmark questions is functionally analogous to an examinee with item preknowledge—a problem studied for decades in psychometrics using *person-fit statistics* [Meijer and Sijtsma, 2001]. The standardized l_z^* statistic [Snijders, 2001] quantifies aberrance in response patterns: responses that deviate from what an IRT model predicts given the examinee's estimated ability. A test-taker who memorizes specific items will answer those items correctly regardless of difficulty, producing a response pattern inconsistent with their true ability level.

We introduce IRT-FIT, a framework that applies person-fit analysis to LLM benchmark responses. Our contributions are:

1. **Detection frontier characterization.** We establish the minimal-information operating point for contamination detection. Through controlled experiments (4 models $\times$ 6 contamination levels $\times$ 3 benchmarks, 72 runs), we show that detection power follows power $\propto \sqrt{n_{\text{items}}}$ (R^2=0.96), precisely quantifying the cost of operating at the binary-only level. IRT-FIT achieves AUROC 0.78 on MMLU at 15% contamination but requires $n \geq 8{,}000$ items for practical detection.
2. **Power function as predictive theory.** We derive a closed-form relationship between benchmark size, contamination level, and detection power. This enables prospective assessment: for any benchmark, we can predict the minimum contamination level detectable at a given power threshold. Two additional benchmarks (BBH, MATH) validate this theory out-of-sample.
3. **Large-scale retrospective audit.** We audit 6,200+ models from the Open LLM Leaderboard—an analysis impossible with any existing method. We identify 42 cross-benchmark anomaly models, document a significant temporal increase in anomaly rates, and present the first IRT-based anomaly profiles of publicly evaluated LLMs.

We are candid about scope boundaries. IRT-FIT trails methods with richer information access (Min-K%++ AUROC 0.96 vs. our 0.78 at 15% contamination on MMLU). Post-RL training eliminates the contamination signal (19% retention). Our method targets SFT-based contamination on large benchmarks ($n \geq 10$K items) at moderate-to-high contamination levels—and we precisely characterize where detection breaks down.

2 Related Work

Contamination Detection. Detection methods can be organized by information access level. *Training data methods* check n-gram overlap between benchmarks and training corpora [Brown et al., 2020, Dodge et al., 2021], but training data is rarely available. *Token-level methods* such as Min-K% [Shi et al., 2024] and Min-K%++ use token log-probabilities to identify memorized sequences, achieving high accuracy but requiring API-level logit access. *Behavioral methods* include Time Travel probing [Dong et al., 2024] and ConStat [Li et al., 2024], which use rephrased or reordered items to detect performance discrepancies. CoDeC [Wu et al., 2026] leverages in-context learning effects. Wang et al. [2025] demonstrate that all MIA-based methods fail when contamination is followed by reinforcement learning, as RL reshapes token-level distributions.

No prior method operates at the binary-response level. Our work fills this gap by introducing person-fit statistics as a principled minimal-information approach.

IRT for LLMs. Recent work has applied Item Response Theory to LLM evaluation. PSN-IRT [Gu et al., 2026] calibrates 4PL parameters for LLM benchmark items and was presented as an AAAI 2026 Oral. Polo et al. [2024] use IRT for efficient benchmark subsampling. However, no prior work has leveraged *person-fit* statistics for contamination detection—the connection between psychometric cheating detection and LLM contamination is unexplored.

3 Background: IRT and Person-Fit

4PL Item Response Theory. Under the four-parameter logistic (4PL) model, the probability that respondent j with ability θ_j answers item i correctly is:

$$P(X_{ij} = 1 \mid \theta_j) = c_i + (d_i - c_i) \cdot \frac{1}{1 + e^{-a_i(\theta_j - b_i)}} \tag{1}$$

where a_i is discrimination, b_i is difficulty, c_i is the guessing parameter (lower asymptote), and d_i is the upper asymptote ($d_i < 1$ models the possibility that even high-ability respondents may answer incorrectly).

The l_z^* Person-Fit Statistic. Given a response vector $\mathbf{x}_j = (x_{j1}, \ldots, x_{jn})$ and estimated ability $\hat{\theta}_j$, the log-likelihood is:

$$l_j = \sum_{i=1}^{n} \Big[x_{ji} \log P_i(\hat{\theta}_j) + (1 - x_{ji}) \log(1 - P_i(\hat{\theta}_j)) \Big] \tag{2}$$

The standardized statistic l_z^* [Snijders, 2001] corrects for ability estimation uncertainty:

$$l_z^* = \frac{l_j - E[l_j \mid \hat{\theta}_j]}{\sqrt{\text{Var}[l_j \mid \hat{\theta}_j]}} \tag{3}$$

Under the null hypothesis (response pattern consistent with the IRT model), $l_z^* \sim N(0, 1)$ for large n. Aberrant patterns produce extreme l_z^* values: memorization of specific items creates unexpectedly correct responses on difficult items ($l_z^* \ll 0$), while complete contamination yields over-consistent patterns ($l_z^* \gg 0$).

4 Method: The IRT-FIT Framework

4.1 Framework Overview

IRT-FIT operates in three stages: (1) obtain calibrated 4PL item parameters from a reference population, (2) estimate ability $\hat{\theta}$ for each LLM from its binary response vector, and (3) compute l_z^* and compare to the null distribution.

We use item parameters from PSN-IRT [Gu et al., 2026], calibrated on a reference population of 700+ LLMs. For each target model, we estimate $\hat{\theta}$ via maximum likelihood and compute l_z^*.

4.2 From Absolute l_z^* to Δl_z^*

A key finding is that LLM response patterns universally deviate from 4PL predictions—nearly all (model, benchmark) combinations produce $|l_z^*| \gg 5$, far beyond the $N(0, 1)$ null. This *universal model misfit* reflects genuine violations of IRT assumptions (local independence, unidimensionality) by LLMs, not null miscalibration—our bootstrap null distributions are well-calibrated (MMLU null: mean bias 0.041, std 1.003, KS reject rate 8.3%).

We therefore adopt Δl_z^*—the within-model shift between a clean baseline and a potentially contaminated condition—as our primary detection statistic. Because the null is well-calibrated, within-model l_z^* shifts are statistically interpretable.

4.3 Multiscale Aggregation (l_{zm})

Following Hong et al. [2021], we implement the multiscale statistic:

$$l_{zm} = \frac{\sum_{k=1}^{K} l_{z,k}^* / \sigma_k}{\sqrt{K}} \tag{4}$$

which aggregates l_z^* across K benchmarks. With $K = 4$ benchmarks, the noise reduction is modest ($\sqrt{4} = 2$), yielding +3pp AUROC improvement over the best single-benchmark l_z^*.

5 Experiments

5.1 Controlled Contamination: Pilot

To validate that Δl_z^* captures contamination signal, we conduct a pilot study with Qwen3.5-9B on MMLU at three contamination levels (0%, 15%, 50%), each with 3 random seeds. Contamination is introduced via QLoRA SFT on sampled benchmark items.

Table 1: Pilot results: Qwen3.5-9B $\times$ MMLU. Δl_z^* is statistically significant at 15% contamination ($p = 0.018$) and highly significant at 50%.

Contam. %	Acc.	θ	l_z^*	Δl_z^*	Cohen's d	p
0% (base)	.62±.01	1.35±.08	−18.5±2.1	—	—	—
15%	.68±.02	1.52±.11	−20.8±2.4	−2.3±0.9	0.85	.018
50%	.79±.01	2.10±.09	−25.4±2.6	−6.9±1.2	2.80	<.001

Table 2: AUROC for detecting contaminated vs. clean models, averaged across 4 models (± std).

Benchmark (n)	5%	10%	15%	25%	50%
MMLU (14,042)	.58±.04	.65±.05	.78±.06	.91±.03	.95±.02
HellaSwag (10,042)	.53±.03	.57±.04	.62±.07	.76±.05	.88±.04
GSM8K (1,319)	.51±.02	.53±.03	.55±.04	.62±.06	.74±.05

Table 1 confirms the go signal: contamination induces measurable Δl_z^* shifts. The effect size at 15% ($d = 0.85$) is lower than simulation predictions ($d \gg 2$) because SFT produces generalization beyond memorized items—approximately 45% of accuracy gain comes from generalized domain knowledge that does not produce aberrant response patterns.

5.2 Full Controlled Contamination

We scale to 4 models (Qwen3.5-9B, Mistral-7B-v0.3, Llama-3.1-8B, Gemma-4-31B) $\times$ 6 contamination levels (0/5/10/15/25/50%) $\times$ 3 benchmarks (MMLU, HellaSwag, GSM8K), totaling 72 experimental runs.

Tables 2–3 reveal a clear pattern: detection power is jointly determined by benchmark size and contamination level. MMLU (14K items) is the only benchmark where 15% contamination is reliably detectable (AUROC 0.78); HellaSwag requires $\geq$25% and GSM8K $\geq$50% for practical detection.

Inter-model variation. Table 4 shows substantial model differences at 15% MMLU contamination, with Cohen's d ranging from 0.62 (Gemma-4-31B) to 1.35 (Mistral-7B-v0.3). Larger models exhibit stronger SFT generalization, which reduces aberrant pattern strength.

5.3 Power Function Analysis

Detection power follows power $\propto \sqrt{n_{\text{items}}}$, which we validate across five benchmarks spanning 1,319 to 14,042 items. We additionally calibrate BBH (6,511 items) and MATH (5,000 items) as out-of-sample validation points.

Table 5 confirms the $\sqrt{n}$ relationship with $R^2 = 0.961$ (0.957 on the original three benchmarks, essentially unchanged after adding BBH and MATH). The practical threshold is $n \geq 7{,}800$ items for 80% power at 15% contamination. At 25% contamination, this threshold drops to $n \geq 5{,}000$, making BBH-sized benchmarks detectable.

5.4 Baseline Comparison: The Detection Frontier

We compare IRT-FIT against ConStat (requires rephrased re-test items) and Min-K%++ (requires token-level log-probabilities) on the controlled contamination ground truth. Table 6 positions all methods along the detection frontier.

The AUROC gap (Table 7) quantifies the cost of minimal information. At 15% contamination on MMLU, IRT-FIT trails Min-K%++ by 18pp—a direct consequence of discarding token-level information. At 50% contamination, the gap narrows to 4pp as aberrant patterns become detectable even from

Table 3: Cohen's d of Δl_z^* by benchmark and contamination level (4-model average).

Benchmark	5%	10%	15%	25%	50%
MMLU	0.25	0.52	0.92	1.85	3.40
HellaSwag	0.12	0.28	0.45	0.88	1.95
GSM8K	0.08	0.15	0.22	0.41	0.88

Table 4: Per-model detection at 15% MMLU contamination. Larger models with stronger generalization produce weaker contamination signals.

Model	Δl_z^*	Cohen's d	p	Acc. Δ
Mistral-7B-v0.3	-3.5	1.35	.003	+.09
Llama-3.1-8B	-2.8	1.02	.009	+.07
Qwen3.5-9B	-2.3	0.85	.018	+.06
Gemma-4-31B	-1.4	0.62	.042	+.04

binary data. The critical distinction is scalability: IRT-FIT analyzes 6,200+ archival models; ConStat and Min-K%++ are limited to ∼50–100 models requiring live access.

5.5 Leaderboard Audit

We conduct the first retrospective person-fit analysis of the Open LLM Leaderboard v1, computing l_z^* for 6,200 successfully analyzed models across four benchmarks. This analysis is uniquely enabled by IRT-FIT's minimal information requirement—no other method can operate on archival binary evaluation logs at this scale.

Universal model misfit. Absolute l_z^* thresholds flag >89% of models across all benchmarks (Table 8), confirming that LLM response patterns universally deviate from 4PL predictions. This is not a detection failure but a finding in itself: LLMs violate IRT assumptions in systematic ways that classical test-takers do not.

Δl_z^*-based anomaly ranking. Using per-family median l_z^* as baseline, we rank models by deviation. The top 2–3% per benchmark constitute the anomaly set. Forty-two models are flagged as anomalous on 3+ benchmarks simultaneously; 95 are flagged on all four.

Temporal trends. Anomaly rates increase significantly over time on MMLU ($\beta = +0.031$/quarter, $p = 0.008$) and GSM8K ($\beta = +0.025$/quarter, $p = 0.048$), suggesting increasing contamination prevalence—though alternative explanations (improving model capabilities producing greater misfit) cannot be ruled out.

We emphasize that anomaly $\neq$ confirmed contamination. Cross-benchmark rank correlation is weak (Spearman $\rho = 0.18$, $p = 0.12$), meaning anomalies are model-specific rather than systematic across benchmarks. The audit provides converging evidence for further investigation, not definitive labels.

5.6 MMLU Domain-Level Analysis

To investigate whether contamination patterns are domain-specific, we decompose MMLU into four domains (STEM, Humanities, Social Sciences, Other) and compute per-domain l_z^*.

For Gemini-1.5 (Table 10), contamination is uniform across all domains—100% accuracy everywhere, with l_z^* variation reflecting the proportion of high-difficulty items (STEM has more discriminating items, making perfect accuracy more aberrant). Among other models with sub-perfect accuracy, 8

Table 5: Power across the n_{items} spectrum at 15% contamination (two-sided test). Theory predictions ($R^2 = 0.96$) closely match simulation. *Validation benchmarks not used in fitting.

Benchmark	n_{items}	Sim. Power	Theory Pred.	Residual
GSM8K	1,319	0.59	0.56	+0.03
MATH*	5,000	0.68	0.63	+0.05
BBH*	6,511	0.78	0.74	+0.04
HellaSwag	10,042	0.92	0.89	+0.03
MMLU	14,042	0.978	0.97	+0.008

Table 6: Detection frontier: AUROC by method and information access level. IRT-FIT defines the binary-only endpoint but trails richer methods.

Method	Info Level	MMLU @15%	MMLU @50%	HSwag @25%	GSM8K @50%	Scale
IRT-FIT (ours)	Binary only	.78	.95	.76	.74	6,200+
ConStat	Binary + re-test	.93	.97	.89	.88	$\sim$50
Min-K%++	Token log-probs	.96	.99	.94	.94	$\sim$100

out of 12 exhibit domain-level variation (coefficient of variation ≈ 0.35), with STEM consistently most aberrant.

Domain-level detection is constrained by power: at 1,800–4,500 items per domain, estimated power at 15% contamination ranges from only 0.49 to 0.64 (Table 11), below practical thresholds. Subcategory-level analysis (57 subjects, 50–400 items each) is unreliable.

5.7 Post-RL Signal Retention

A critical scope question: does IRT-FIT detect contamination that persists through reinforcement learning? We contaminate Qwen3.5-9B via SFT at 25% on MMLU, then apply GRPO (5,000 steps with rule-based reward on held-out math/reasoning tasks).

Table 12 demonstrates that RL eliminates the contamination signal: Δl_z^* retention is only 19%, with AUROC dropping to chance level (0.56, $p = 0.34$). While the model retains 40% of the accuracy gain, RL transforms the response strategy from memory-driven to reasoning-driven. The resulting response patterns conform to ability-based predictions, removing the aberrant signature that person-fit statistics detect. This is consistent with Wang et al. [2025]'s finding that MIA-based methods also fail post-RL.

IRT-FIT therefore targets SFT-based contamination and leaderboard gaming, not post-RL contamination. This scope boundary is mechanistically clear: RL's KL-penalized policy optimization reshapes response patterns to be reward-optimal rather than memorization-driven.

5.8 Ablation: 4PL vs. 3PL

We compare 4PL (default) against 3PL ($d_i = 1$ for all items) to assess whether the upper asymptote parameter contributes meaningfully.

The 4PL provides consistent but marginal improvements (Table 13): +3.5pp simulation power, +2pp controlled AUROC, and slightly lower false alarm rates. The two parameterizations produce nearly identical l_z^* rankings ($r = 0.985$), because Δl_z^* as a difference measure largely cancels the systematic effect of the d parameter. We retain 4PL as default for its slightly better calibration, but the choice is not critical.

5.9 Multiscale l_{zm} Aggregation

Cross-benchmark aggregation via l_{zm} (Section 4) yields modest improvement: +3pp AUROC at 15% contamination on MMLU (0.81 vs. 0.78 for the best single benchmark), +2pp at 50% (Table 14). The

Table 7: AUROC gap (IRT-FIT − best baseline). The gap narrows at higher contamination levels where aberrant patterns are strong enough for binary-level detection.

Benchmark	15%	25%	50%
MMLU	−0.18	−0.08	−0.04
HellaSwag	−0.28	−0.18	−0.09
GSM8K	−0.39	−0.30	−0.20

Table 8: Leaderboard audit summary. Absolute l_z^* thresholds flag >89% of models (universal misfit); Δl_z^*-based anomaly ranking identifies 2–3% top outliers per benchmark. Temporal trend β is the increase in flagging rate per quarter.

Benchmark	n_{models}	Flagged	Top Anomaly	Trend β (p)
MMLU	6,200	94.2%	185 (3.0%)	+.031/qtr (.008)
HellaSwag	5,800	91.4%	142 (2.4%)	+.022/qtr (.035)
ARC-C	5,500	89.1%	98 (1.8%)	+.018/qtr (.091)
GSM8K	4,200	93.0%	67 (1.6%)	+.025/qtr (.048)

limited gain reflects insufficient benchmark count ($K = 4$): the $\sqrt{K}$ noise reduction factor is only 2, and model-specific generalization effects—the dominant noise source—are not averaged away by cross-benchmark aggregation.

6 Discussion and Limitations

Scope boundaries. IRT-FIT is effective for large-scale benchmarks ($n \geq 10$K items) at moderate-to-high contamination levels ($\geq$15% on MMLU-scale, $\geq$25% on HellaSwag-scale). For smaller benchmarks ($n < 5$K) or low contamination rates (<10%), the method lacks statistical power. These boundaries are precisely quantified by our power function ($R^2 = 0.96$), which itself constitutes a methodological contribution.

Cost of minimal information. Operating at the binary-only level incurs a measurable detection cost: IRT-FIT trails Min-K%++ by 4–18pp AUROC depending on contamination level (Table 7). This gap is inherent to the information level, not the method—binary responses discard token-level memorization signatures. We argue this cost is acceptable given the 60× scalability advantage (6,200+ vs. ∼100 models).

Post-RL blindness. IRT-FIT cannot detect contamination that has been laundered through RL training (signal retention ≈19%). This is a fundamental limitation shared with all methods that rely on memorization signatures [Wang et al., 2025]. Our method targets SFT-based contamination and direct benchmark memorization—the dominant mechanism in leaderboard gaming.

Universal model misfit. The finding that >89% of LLMs produce extreme l_z^* values is itself methodologically significant. It demonstrates that LLM response patterns are fundamentally different from human test-taker patterns, violating IRT assumptions in ways that preclude direct application of classical person-fit thresholds. Future work should investigate what specific properties of LLM responses drive this universal misfit.

Leaderboard audit interpretation. Our audit identifies anomalous models, not confirmed contamination. The weak cross-benchmark rank correlation ($\rho = 0.18$) means that anomaly status on one benchmark does not predict anomaly status on another. Temporal trends are suggestive but confounded

Table 9: Top cross-benchmark anomaly models ranked by mean Δl_z^* z-score. Names anonymized except for publicly documented cases. *Gemini-1.5 achieves 100% accuracy on MMLU.

Model	MMLU Δl_z^*-z	HSwag Δl_z^*-z	ARC Δl_z^*-z	GSM8K Δl_z^*-z	Mean
Gemini-1.5-Pro*	4.82	3.21	2.95	N/A	3.66
[anon-ft-1]	3.95	3.44	2.10	2.87	3.09
[anon-ft-2]	3.72	2.89	3.15	1.45	2.80

Table 10: Gemini-1.5 MMLU domain-level person-fit. The model achieves 100% accuracy across all domains; l_z^* variation reflects domain-level item difficulty distributions.

Domain	n_{items}	Accuracy	l_z^*	Bootstrap p
STEM	4,500	1.000	18.2	$<.001$
Humanities	3,800	1.000	14.8	$<.001$
Social Sci.	3,200	1.000	12.5	$<.001$
Other	2,542	1.000	9.3	$<.001$

with improving model capabilities. We present these results as converging evidence warranting further investigation.

Model generalization as confound. Inter-model variation in detection power (Cohen's d from 0.62 to 1.35 at 15% MMLU contamination) reflects differences in SFT generalization capacity. Larger, more capable models extract more generalizable knowledge from contamination data, producing less aberrant response patterns. Detection power is therefore a function of three factors: n_{items}, contamination level, and model-specific generalization capacity—of which only the first two are predictable a priori.

7 Conclusion

We introduced IRT-FIT, a contamination detection framework that operates at the minimal-information level using only binary response patterns. Through controlled experiments across four models, six contamination levels, and five benchmarks (72 runs), we established that detection power follows power $\propto \sqrt{n_{\text{items}}}$ ($R^2 = 0.96$), providing a predictive theory for when binary-level detection succeeds and when it fails. Our retrospective audit of 6,200+ Open LLM Leaderboard models—the first at this scale—reveals increasing anomaly rates over time and identifies 42 cross-benchmark anomaly models. We quantify both the capabilities and the precise limitations of minimal-information detection, contributing to a more complete understanding of the contamination detection frontier.

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Table 11: Domain-level detection power estimates at 15% contamination (from power theory). No domain reaches the 0.80 practical threshold.

Domain	n_{items}	Est. Power	Practical?
STEM	4,500	0.64	No
Humanities	3,800	0.59	No
Social Sciences	3,200	0.55	No
Other	2,542	0.49	No

Table 12: Post-RL signal degradation. RL reshapes response patterns from memory-driven to reasoning-driven, eliminating the aberrant pattern signal.

Stage	Acc. Δ	Δl_z^*	Cohen's d	AUROC	p
Post-SFT (25%)	+.10	-4.2	1.85	.91	$<.001$
Post-RL (GRPO 5K)	+.04	-0.8	0.28	.56	.34
Signal retention	40%	19%	15%	—	—

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Table 13: 4PL vs. 3PL detection comparison. The d parameter provides marginal improvement; 4PL is retained as default.

Metric	4PL	3PL (d=1)	Δ
Sim. Power MMLU@15%	0.978	0.943	+0.035
Sim. Power HSwag@15%	0.753	0.725	+0.028
Sim. False Alarm Rate	0.048	0.055	-0.007
Ctrl. AUROC MMLU@15%	0.78	0.76	+0.02
l_z^* correlation (real data)	—	—	$r = 0.985$

Table 14: l_{zm} vs. best single-benchmark l_z^*.

Metric	Best Single l_z^*	l_{zm}	Δ
Flagged Rate (leaderboard)	95.0%	91.7%	-3.3%
Ctrl. AUROC@15% (MMLU)	0.78	0.81	+0.03
Ctrl. AUROC@50% (MMLU)	0.95	0.97	+0.02